

ARKYA CHATTERJEE

www.mit.edu/~achatt/

EDUCATION

Massachusetts Institute of Technology (MIT)
Ph.D. in Condensed Matter Theory, Department of Physics

May 2025 (expected)

Indian Institute of Technology Bombay (IIT-B)
B.Tech. (with Honors) in Engineering Physics and Minor in Mathematics

May 2019

PUBLICATIONS AND PREPRINTS

A. Chatterjee, S. D. Pace, S.-H. Shao, *Quantized axial charge of staggered fermions and the chiral anomaly*, [arXiv: 2409.12220](#)

Z. D. Shi, **A. Chatterjee**, *Analytic framework for self-dual criticality in $\mathbb{Z}_k$ gauge theory with matter*, [arXiv:2407.07941](#) (submitted)

A. Chatterjee, Ö. M. Aksoy, X.-G. Wen, *Gapped Phases and Phase Transitions in Spin Chains with Noninvertible Symmetries*, [arXiv:2405.05331](#) (submitted)

I. T. Rosen, S. Muschinske, C. N. Barrett, **A. Chatterjee**, *et al.*, *Implementing a synthetic magnetic vector potential in a 2D superconducting qubit array*, [arXiv:2405.00873](#) (accepted in Nature Physics)

A. Chatterjee, W. Ji, X.-G. Wen, *Emergent Generalized Symmetry and Maximal Symmetry-Topological-Order*, [arXiv:2212.14432](#) (submitted)

A. Chatterjee, X.-G. Wen, *Holographic Theory for Continuous Phase Transitions: Emergence and symmetry protection of gaplessness*, *Phys. Rev. B* **108**, 075105 (2023)

A. Chatterjee, X.-G. Wen, *Symmetry as a Shadow of Topological Order and a Derivation of Topological Holographic Principle*, *Phys. Rev. B* **107**, 155136 (2023)

M. Chatterjee, **A. Chatterjee**, A. Nandi, A. Sain, *Dynamics and stability of contractile actomyosin ring in the cell*, *Phys. Rev. Lett.* **128**, 068102 (2022)

A. Fischer, **A. Chatterjee**, T. Speck, *Aggregation and sedimentation of active Brownian particles at constant affinity*, *J. Chem. Phys.* **150**, 064910 (2019)

INVITED TALKS

[Condensed Matter Seminar](#) (virtual), National University of Singapore

November 2024

Quantum Matter in Mathematics & Physics seminar

June 2024

Harvard University Center of Mathematical Sciences and Applications

[Symmetry Seminar](#) (virtual)

May 2024

Condensed Matter Theory (CMT) seminar, Indian Institute of Science

April 2024

CMT Kids' seminar, Harvard University

February 2024

CMT seminar, Boston University

November 2023

Physics colloquium, IIT-B

June 2023

CMT seminar (virtual), Pennsylvania State University

April 2023

CONFERENCES AND SUMMER SCHOOLS

Paths to Quantum Field Theory 2024: short talk University of Sarajevo	July 2024
Prospects in Theoretical Physics 2024 summer school: poster & gong show Institute for Advanced Study	July 2024
Summer school on “Symmetries and Anomalies: a modern take”: poster & gong show Institut des Hautes Études Scientifiques	June 2024
Higher Categorical Tools for Quantum Phases of Matter: gong show Perimeter Institute for Theoretical Physics	March 2024
Simons Ultra-Quantum Matter collaboration meeting: poster University of Colorado Boulder	May 2023

SELECTED HONORS AND FELLOWSHIPS

Invitation to the KITP program on “Generalized Symmetries in Quantum Field Theory”	2025
C. M. Clay and H. W. Kendall Fellowship, MIT	2019
K. Seshia Research Excellence Award for best undergraduate thesis, IIT-B Physics	2019
DAAD WISE fellowship for a research internship in Germany	2018
Institute Academic Prize for highest GPA in the Physics department, IIT-B	2018

TEACHING AND MENTORING

Massachusetts Institute of Technology, Cambridge, MA

- Teaching Assistant for quantum, statistical, and classical mechanics classes at undergraduate & graduate levels; conducted weekly recitations and contributed to setting and grading exams.
- Mentored undergraduate students as part of the Physics Directed Reading Program

Indian Institute of Technology Bombay, Mumbai, India

- Teaching Assistant for first-year undergraduate electromagnetism and calculus classes
- Mentor for the Summer of Science reading projects

SERVICE AND OUTREACH

Project SHORT , Mentor	2024 – present
Responsible for providing 1-on-1 graduate school application assistance to prospective Physics PhD candidates; aimed at shrinking the socioeconomic gap in graduate school	
SciPost Physics and Physical Review B , Referee	2023 – present
Refereed articles on condensed matter and quantum information physics	
MIT Physics Graduate Students Council , Webmaster	2020 – 22
Responsible for maintaining the PGSC website, providing web-related support to the Physics Values Committee	

- Harvard Museum of Natural History**, Volunteer *2022*
Gallery facilitator for Glass Flowers, Maya Civilization, and Sharks exhibits
- IIT-B Maths & Physics Club**, Invited Speaker *2020*
Gave a (virtual) pedagogical talk to an undergraduate audience on the quantum Hall effect
- IIT-B Physics Academic Mentorship Program**, Co-Lead *2018 – 19*
Led a team of 12 mentors to provide academic support and facilitate outreach to ~ 200 students